

Digital Signal Processing

Detailed Exam Solutions (Part II)

Generated Solution Document

Introduction

This document provides complete, step-by-step solutions for all questions in **PART II** of the Digital Signal Processing exam paper (Ref No: Ex/EE/PC/B/T/224/2025).

Question 1

Main Question: DFT and Frequency Samples

Problem: Derive the expression of DFT of a sequence of length N . Hence show that the number of time-samples is equal to the number of frequency samples.

Solution:

The Discrete Fourier Transform (DFT) of a finite-length sequence $x[n]$ of length N (where $0 \leq n \leq N-1$) is defined by sampling its Discrete-Time Fourier Transform (DTFT) $X(e^{j\omega})$ at N equally spaced frequencies $\omega_k = \frac{2\pi k}{N}$ for $k = 0, 1, \dots, N-1$. The DTFT is given by:

$$X(e^{j\omega}) = \sum_{n=0}^{N-1} x[n]e^{-j\omega n}$$

Substituting $\omega = \frac{2\pi k}{N}$, we get the DFT sequence $X[k]$:

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}kn} = \sum_{n=0}^{N-1} x[n]W_N^{kn} \quad \text{for } k = 0, 1, \dots, N-1$$

where $W_N = e^{-j2\pi/N}$ is the N -th root of unity (Twiddle factor).

Number of samples: The sequence $x[n]$ is defined for $n = 0, 1, \dots, N-1$, meaning there are exactly N time-domain samples. In the frequency domain, because the complex exponential W_N^{kn} is periodic in k with period N ($W_N^{(k+N)n} = W_N^{kn}W_N^{Nn} = W_N^{kn}(1) = W_N^{kn}$), the DFT sequence $X[k]$ is also periodic with period N . Therefore, there are exactly N unique frequency-domain samples (from $k = 0$ to $N-1$). Thus, the number of time-samples (N) strictly equals the number of frequency samples (N).

OR Question (a): FFT Computational Savings

Problem: Prove that the algorithm of FFT reduces the computational burden. Estimate the savings.

Solution:

To compute an N -point DFT directly using the defining formula, for each of the N values of k ,

we must compute a sum of N terms. Each term requires a complex multiplication and a complex addition. Therefore, the direct DFT requires approximately N^2 complex multiplications and $N(N-1)$ complex additions. The computational complexity is $\mathcal{O}(N^2)$.

The Fast Fourier Transform (FFT), such as the Radix-2 Cooley-Tukey algorithm, takes advantage of the symmetry and periodicity of the twiddle factors ($W_N^{k+N/2} = -W_N^k$). It divides the N -point DFT into two $(N/2)$ -point DFTs, applying this recursively. The number of stages in the decimation process is $\log_2 N$. Each stage involves $N/2$ butterfly operations, and each butterfly requires 1 complex multiplication and 2 complex additions. Thus, the FFT requires:

- Complex multiplications: $\frac{N}{2} \log_2 N$
- Complex additions: $N \log_2 N$

Savings Estimation: The ratio of multiplications for Direct DFT vs. FFT is:

$$\text{Speed-up Factor} = \frac{N^2}{\frac{N}{2} \log_2 N} = \frac{2N}{\log_2 N}$$

For large N , this ratio is massive, proving the immense reduction in computational burden.

OR Question (b): Twiddle Factor Property

Problem: "If W is the Twiddle Factor used to perform N -point DFT operation, then always $W^{pN/2} = (-1)^p$, p being an integer." Comment if true or false and justify.

Solution:

The twiddle factor for an N -point DFT is $W_N = e^{-j\frac{2\pi}{N}}$. Let us evaluate $W_N^{pN/2}$:

$$W_N^{pN/2} = \left(e^{-j\frac{2\pi}{N}} \right)^{\frac{pN}{2}} = e^{-j\frac{2\pi}{N} \cdot \frac{pN}{2}} = e^{-j\pi p}$$

Using Euler's identity, $e^{-j\pi p} = \cos(\pi p) - j \sin(\pi p)$. Since p is an integer, $\sin(\pi p) = 0$ and $\cos(\pi p) = (-1)^p$. Therefore, $W_N^{pN/2} = (-1)^p$.

Conclusion: The statement is **TRUE**.

Question 2

Main (a) & (b): Linear Phase Symmetry and Distortion-less Filter

(a) Symmetry of magnitude and phase spectra:

For a linear phase digital filter with a real-valued impulse response $h[n]$, the frequency response is $H(e^{j\omega}) = \sum h[n]e^{-j\omega n}$. Since $h[n]$ is real, $H(e^{-j\omega}) = H^*(e^{j\omega})$. Expressing in polar form $H(e^{j\omega}) = |H(e^{j\omega})|e^{j\theta(\omega)}$, we have:

$$|H(e^{-j\omega})|e^{j\theta(-\omega)} = |H(e^{j\omega})|e^{-j\theta(\omega)}$$

Equating magnitude and phase components:

- Magnitude is **even**: $|H(e^{j\omega})| = |H(e^{-j\omega})|$
- Phase is **odd**: $\theta(-\omega) = -\theta(\omega)$

Because the response is discrete-time, it is periodic with 2π . The symmetry reflects around $\omega = \pi$ (and 0). The frequency $\omega = \pi$ rad/sample is the **folding frequency** (or Nyquist frequency).

Relation to Sampling Frequency (F_s): The mapping is $\omega = 2\pi \frac{F}{F_s}$. Setting $\omega = \pi$, we find the folding frequency $F_{fold} = \frac{F_s}{2}$.

(b) True/False Justification: "A distortion-less filter has constant phase delay in its pass band region."

Conclusion: TRUE.

Justification: A distortion-less transmission requires the output to be an exact replica of the input, possibly scaled and delayed: $y(t) = Kx(t - t_d)$. In the frequency domain, this requires $H(\omega) = Ke^{-j\omega t_d}$. This means: 1. Magnitude $|H(\omega)| = K$ (constant). 2. Phase $\theta(\omega) = -\omega t_d$ (linear phase). Phase delay is defined as $\tau_p = -\frac{\theta(\omega)}{\omega} = -\frac{-\omega t_d}{\omega} = t_d$. Since t_d is a constant time delay, the phase delay must be constant in the passband.

OR (a) & (b): Brick-wall FIR filter problems & Tapped Delay

(a) Brick-wall FIR design problems:

Designing an ideal brick-wall FIR filter requires an impulse response $h_d[n]$ that forms a *sinc* function extending from $-\infty$ to ∞ . Since a practical FIR filter must be finite, we must truncate $h_d[n]$. **Problem:** Abrupt truncation causes the **Gibbs Phenomenon**, which manifests as large, non-diminishing oscillatory ripples in both the passband and stopband near the cut-off frequency. **Measures:** To overcome this, the truncated impulse response is multiplied by a smooth tapering function called a **Window** (e.g., Hamming, Hann, Blackman window). This gradually attenuates the ends of the impulse response sequence to zero. **Achievement:** We **cannot** achieve the exact desired "brick-wall" response. Windowing suppresses the ripples, but it comes at the cost of widening the transition band (it is no longer perfectly sharp/vertical).

(b) Tapped delay structure Computational load:

An M -tap causal FIR filter difference equation is $y[n] = \sum_{k=0}^{M-1} h[k]x[n-k]$. It requires a tapped delay line consisting of:

- $M - 1$ delay elements (z^{-1})
- M multipliers (for coefficients $h[0]$ to $h[M - 1]$)
- $M - 1$ adders

Computational load: For every single output sample $y[n]$, the processor must execute M Multiply-and-Accumulate (MAC) operations. The load is directly proportional to the filter length M , making it $\mathcal{O}(M)$ per sample.

Question 3

Main: FIR Frequency Response Group Delay

(a) Derivation for symmetric FIR:

Let the causal FIR filter have length M and symmetric coefficients $h[n] = h[M - 1 - n]$. The frequency response is $H(e^{j\omega}) = \sum_{n=0}^{M-1} h[n]e^{-j\omega n}$. We can factor out the center delay term $e^{-j\omega \frac{M-1}{2}}$:

$$H(e^{j\omega}) = e^{-j\omega \frac{M-1}{2}} \sum_{n=0}^{M-1} h[n]e^{j\omega(\frac{M-1}{2}-n)}$$

Due to symmetry $h[n] = h[M - 1 - n]$, terms at index n and $M - 1 - n$ can be paired. Let M be odd (Type I), the center tap is at $\alpha = (M - 1)/2$.

$$H(e^{j\omega}) = e^{-j\omega\alpha} \left[h[\alpha] + \sum_{n=0}^{\alpha-1} 2h[n] \cos(\omega(\alpha - n)) \right]$$

This shows $H(e^{j\omega}) = H_r(\omega)e^{-j\omega\alpha}$, where $H_r(\omega)$ is a strictly real amplitude function.

(b) Group Delay:

Group delay $\tau_g = -\frac{d\theta(\omega)}{d\omega}$. Since the phase is strictly $\theta(\omega) = -\omega \left(\frac{M-1}{2} \right)$ (plus possible phase jumps of π where $H_r(\omega)$ changes sign), the group delay is strictly constant:

$$\tau_g = \frac{M-1}{2} \text{ samples}$$

(c) Magnitude and Phase Symmetry:

As shown in Q2(a), for any real impulse response, $|H(e^{j\omega})| = |H(e^{-j\omega})|$ (Magnitude is Even), and $\theta(\omega) = -\theta(-\omega)$ (Phase is Odd). The explicit derivation above also shows $|H(e^{j\omega})| = |H_r(\omega)|$, which consists of cosine terms that are even functions, confirming magnitude evenness. The phase is $-\omega\alpha$, which is directly a linear, odd function.

OR: Applications of DFT Butterfly Operation

(a) Applications of DFT: 1. **Spectral Analysis:** Using the DFT/FFT to extract the frequency components (spectrum) of a finite discrete-time signal. It helps in identifying dominant frequencies, harmonics, and noise in audio, radar, and biomedical signals. 2. **Fast Convolution / Filtering:** Computing the linear convolution of two sequences via the frequency domain. Using the property that time-domain circular convolution is frequency-domain multiplication, sequences are zero-padded and their DFTs are multiplied, then IDFT is taken. It drastically speeds up filtering operations for long sequences.

(b) Butterfly Mathematical Operations: For a 2-input radix-2 decimation-in-time butterfly, let the inputs from the current stage m be A and B , and the twiddle factor be W . The outputs for stage $m+1$ are $X_{m+1}[p]$ and $X_{m+1}[q]$:

$$\begin{aligned} X_{m+1}[p] &= A + B \cdot W \\ X_{m+1}[q] &= A - B \cdot W \end{aligned}$$

Sequence of Operations: 1. Fetch B and W . 2. Multiply $B \times W$ (One complex multiplication). 3. Fetch A . 4. Add $(B \times W)$ to A to form the upper node output $X_{m+1}[p]$ (One complex addition). 5. Subtract $(B \times W)$ from A to form the lower node output $X_{m+1}[q]$ (One complex subtraction/addition).

Question 4 (Answer ONE)

(a) 7-tap FIR Hann Window Design

Problem: 7-tap causal linear-phase FIR brick-wall LP filter. Gain=10, $f_c = 500$ Hz, $F_s = 2000$ Hz. Apply Hann window.

Solution:

1. **Filter Specifications:** $M = 7$, $\alpha = 3$. Cut-off in rad: $\omega_c = 2\pi(500)/2000 = \pi/2$. Gain

$G = 10$. 2. **Ideal Coefficients** $h_d[n]$:

$$h_d[n] = G \frac{\sin(\omega_c(n-3))}{\pi(n-3)}$$

- $h_d[3] = G \frac{\omega_c}{\pi} = 10(\frac{1}{2}) = 5$
- $h_d[2] = h_d[4] = 10 \frac{\sin(-\pi/2)}{-\pi} = \frac{10}{\pi} \approx 3.183$
- $h_d[1] = h_d[5] = 10 \frac{\sin(-\pi)}{-2\pi} = 0$
- $h_d[0] = h_d[6] = 10 \frac{\sin(-3\pi/2)}{-3\pi} = -1.061$

3. **Hann Window** $w[n]$: $w[n] = 0.5 - 0.5 \cos(\frac{2\pi n}{M-1}) = 0.5 - 0.5 \cos(\frac{\pi n}{3})$.

- $w[0] = w[6] = 0.5 - 0.5 = 0$
- $w[1] = w[5] = 0.5 - 0.5(0.5) = 0.25$
- $w[2] = w[4] = 0.5 - 0.5(-0.5) = 0.75$
- $w[3] = 0.5 - 0.5(-1) = 1$

4. **Final Coefficients** $h[n] = h_d[n]w[n]$:

- $h[0] = h[6] = -1.061 \times 0 = 0$
- $h[1] = h[5] = 0 \times 0.25 = 0$
- $h[2] = h[4] = 3.183 \times 0.75 = 2.387$
- $h[3] = 5 \times 1 = 5$

Realization: $\{0, 0, 2.387, 5, 2.387, 0, 0\}$. The filter effectively becomes a 3-tap filter due to the zeros. The tapped delay structure requires only 2 delay elements and 3 multipliers for the non-zero coefficients.

(b) 7-tap Hamming Window Frequency Response

Problem: Frequency response of 7-tap causal Hamming window at $F_s = 1$ kHz. Approximate width of main lobe?

Solution:

For a causal Hamming window of length $M = 7$, the time-domain sequence is:

$$w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{M-1}\right), \quad 0 \leq n \leq M-1$$

In frequency domain, $w[n]$ is the sum of a constant and a cosine over a finite rectangular window of length M . The DTFT of an M -point rectangular window shifted to center at $(M-1)/2$ is:

$$D(\omega) = e^{-j\omega \frac{M-1}{2}} \frac{\sin(\omega M/2)}{\sin(\omega/2)}$$

Using the modulation property, the cosine term splits into two shifted copies of $D(\omega)$:

$$W(e^{j\omega}) = 0.54D(\omega) - 0.23 \left[D\left(\omega - \frac{2\pi}{M-1}\right) + D\left(\omega + \frac{2\pi}{M-1}\right) \right]$$

Main Lobe Width: The theoretical width of the main lobe for a rectangular window is $4\pi/M$. For a Hamming window, the interference of the shifted Dirichlet functions widens the main lobe to exactly $8\pi/M$ rad/sample. With $M = 7$, Main Lobe Width $\Delta\omega = \frac{8\pi}{7}$ rad/sample. Converting to continuous frequency f using $\omega = 2\pi f/F_s$:

$$\Delta f = \frac{8\pi/7}{2\pi} F_s = \frac{4}{7} F_s = \frac{4 \times 1000}{7} \approx 571.4 \text{ Hz}$$

Question 5 (Short Notes)

(a) Circular Convolution and its application:

Circular convolution is a discrete convolution operation defined for periodic sequences or finite-length sequences evaluated on a circle. For two N -point sequences $x_1[n]$ and $x_2[n]$, it is defined as $x_1[n] \otimes_N x_2[n] = \sum_{k=0}^{N-1} x_1[k]x_2[(n-k) \bmod N]$. **Application:** It is fundamentally tied to the DFT: multiplying the DFTs of two sequences yields the DFT of their circular convolution. Because standard linear filtering uses linear convolution, circular convolution must be carefully managed using zero-padding (like overlap-add or overlap-save methods) to ensure it exactly replicates linear convolution, allowing fast filtering via the FFT algorithm.

(b) Distortion-less transmission of signal using FIR digital filter:

A distortion-less system requires constant magnitude and linear phase across the frequency band of interest. FIR filters, particularly when designed with symmetrical (or anti-symmetrical) impulse responses, inherently possess exactly linear phase characteristics. This means all frequency components of a signal passing through the filter experience the exact same time delay (constant group delay). Unlike IIR filters which have non-linear phase leading to phase distortion (dispersion), FIR filters perfectly preserve the waveshape of signals in the passband, making them ideal for applications requiring pristine phase integrity, such as audio processing, data transmission, and biomedical signal analysis.